Design Calculation Sheet

Inputs

Parameter	Value	Unit	Source
Electrode height [m]	0.065		Chen2020 + params_r1_v4.json
Electrode width [m]	1.58		Chen2020 + params_r1_v4.json
Nominal cell capacity [A.h]	3.4		Chen2020 + params_r1_v4.json
Upper voltage cut-off [V]	4.2		Chen2020 + params_r1_v4.json
Lower voltage cut-off [V]	2.5		Chen2020 + params_r1_v4.json
Positive electrode thickness [m]	5.1408e-05		Chen2020 + params_r1_v4.json
Negative electrode thickness [m]	5.7936e-05		Chen2020 + params_r1_v4.json
Separator thickness [m]	1.2e-05		Chen2020 + params_r1_v4.json
Positive current collector thickness [m]	8e-06		Chen2020 + params_r1_v4.json
Negative current collector thickness [m]	6e-06		Chen2020 + params_r1_v4.json
Positive electrode porosity	0.335		Chen2020 + params_r1_v4.json
Negative electrode porosity	0.25		Chen2020 + params_r1_v4.json
Separator porosity	0.55		Chen2020 + params_r1_v4.json
Positive electrode density [kg.m-3]	3262.0		Chen2020 + params_r1_v4.json
Negative electrode density [kg.m-3]	1657.0		Chen2020 + params_r1_v4.json
Separator density [kg.m-3]	397.0		Chen2020 + params_r1_v4.json
Positive current collector density [kg.m-3]	2700.0		Chen2020 + params_r1_v4.json
Negative current collector density [kg.m-3]	8960.0		Chen2020 + params_r1_v4.json
Positive particle radius [m]	1e-06		Chen2020 + params_r1_v4.json
Negative particle radius [m]	1.5e-06		Chen2020 + params_r1_v4.json
Cation transference number	0.5		Chen2020 + params_r1_v4.json
Cell cooling surface area [m2]	0.0075		Chen2020 + params_r1_v4.json
Cell volume [m3]	2.42e-05		Chen2020 + params_r1_v4.json
Total heat transfer coefficient [W.m-2.K-1]	10.0		Chen2020 + params_r1_v4.json
Positive electrode active material volume fraction	0.665		Chen2020 + params_r1_v4.json
Negative electrode active material volume fraction	0.75		Chen2020 + params_r1_v4.json
Positive electrode conductivity [S.m-1]	0.18		Chen2020 + params_r1_v4.json
Negative electrode conductivity [S.m-1]	215.0		Chen2020 + params_r1_v4.json
Positive electrode diffusivity [m2.s-1]	4e-15		Chen2020 + params_r1_v4.json
Negative electrode diffusivity [m2.s-1]	3.3e-14		Chen2020 + params_r1_v4.json
Initial concentration in positive electrode [mol.m-3]	17038.0		Chen2020 + params_r1_v4.json
Initial concentration in negative electrode [mol.m-3]	29866.0		Chen2020 + params_r1_v4.json
pos_cmax	63104.0		Chen2020 + params_r1_v4.json
neg_cmax	33133.0		Chen2020 + params_r1_v4.json

Capacity & Energy

Item	Value	Unit	Source
1C discharge capacity	3.4519	Ah	cell/r1_v4_1c.json
5C discharge capacity	3.4184	Ah	cell/r1_v4_5c.json
5C retention	0.9903	ratio	5C/1C (mechanical)
Discharge energy	12.6797	Wh	∫V·I dt / 3600, cell/r1_v4_energy.json
Midpoint voltage	3.9087	V	cell/r1_v4_energy.json
DC resistance	1.3830	mΩ	cell/r1_v4_energy.json

Energy Density

Item	Value	Unit	Source
Mass (contract)	26.81	g	Σ layer×(1−ε)×ρ×A, cell/r1_v4_energy.json
Energy density	473.01	Wh/kg	energy/mass, cell/r1_v4_energy.json
Volumetric energy density	912.22	Wh/L	cell/r1_v4_energy.json
Stack thickness	135.34	µm	cell/r1_v4_energy.json

N-P & Mass

Item	Value	Unit	Source
N/P ratio	0.765	–	literature stoich windows dx_pos=0.75 (NMC811 0.99->0.24), dx_neg=0.86 (graphite->LiC6); approximate, set does not expose stoich-limit keys
layer positive_electrode	0.11152	kg/m²	cell/r1_v4_energy.json
layer negative_electrode	0.07200	kg/m²	cell/r1_v4_energy.json
layer positive_cc	0.02160	kg/m²	cell/r1_v4_energy.json
layer negative_cc	0.05376	kg/m²	cell/r1_v4_energy.json
layer separator	0.00214	kg/m²	cell/r1_v4_energy.json

Process

Parameter	Value	Unit	Formula
Areal density positive	111.52	g/m²	L×(1−ε)×ρ
Areal density negative	72.00	g/m²	L×(1−ε)×ρ
Compaction density positive	2.1692	g/cm³	ρ×(1−ε)÷1000
Compaction density negative	1.2428	g/cm³	ρ×(1−ε)÷1000
Electrolyte fill	3.93	mL	pore volume; mass @1.2 g/cm³ literature
Formation	0.1C CC to 4.2 V, 25 °C, 2 cycles	–	design recommendation; production value needs tuning